

The Carrier Holds, the Emitter Leaks: A Goal Packet on Its Own Cadence at Creature Scale

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Abstract

The self-bridge results drew a hard boundary: a 256-float packet written by one pass and read by a fresh context can steer what a frozen model *does* (the relay gate, 9/9 at three seeds), but cannot make a small decode *initiate recall* (paper 10, 0.000 across every manipulation tested). A goal sits on the steerable side of that line — a goal-conditioned turn is pure presence, “be the creature currently pursuing X” — and this note walks the mechanism down to creature scale and then around its own loop. C0: the relay gate transplants to a frozen LFM2.5-230M and passes 9 of 9 checks at all three seeds, but only after a registered amendment moves the tap one layer deeper — at the depth-proportional layer the packet memorizes phrasing identity (probe 0.92–1.00 on train against 0.12–0.19 held out, chance), and a per-layer probe scan finds phrasing-invariant content crystallizing exactly at the last full-attention layer. At the bottom of the scale ladder the transplant rule breaks, and the scan is the tap-finder. C1: with the production creature’s self block armed, the packet drives action — goal-consistent tool choice 0.90/0.70/0.90 on held-out phrasings against three in-band floors (no packet 0.20, undirected waking 0.20, wrong-goal packet 0.30–0.50) — and the wrong-goal floor separates *waking* from *directing*: an off-goal packet still elicits purposeful action, aimed wrong. C2 cycles the packet for sixteen ticks with nothing else crossing the pass boundary, and returns the registration’s reading (ii) unanimously: the frozen-packet arm stays goal-consistent at 0.80–1.00 flat across all sixteen ticks — the first measurement of a cycled packet’s lifetime, and the carrier does not decay — while the live re-emitted arm loses the goal on its very first re-encoding: the emitter, trained on assignment exchanges, writes an off-distribution packet the moment it reads the model’s own acted pass, and the arm sits at chance from hop 1 on; the hop-8 goal switch replays the same failure in miniature, one consistent tick and gone. The split licenses a near-term architecture — a per-goal packet store plus a scripted update rule uses only the parts that hold — and names the narrow gap between here and a creature that re-writes its own intention.

1 The boundary that makes a want carryable

Paper 10 ended with a precise negative: at 4B the self-written packet cannot make a free decode initiate recall — twelve (pair, seed) cells, five doses, 0.000 flat — while the relay result it was built on stands untouched: a packet written by one pass, read by a fresh context with no cache and no instruction, message-dependently changes what the model does. The boundary is between *recounting what happened* and *being someone*. A goal needs only the second. “Currently pursuing `listen_pond`” is not an episodic fact to restate; it is a disposition to act, exactly the thing the

channel measurably carries. The null-io line supplies the missing half: `nullin` wakes a frozen model on a null turn from internal state, but cannot say *toward what*. Packet = direction, `nullin` = ignition.

One registered dragon shaped the whole program: nobody had ever *cycled* the packet. The relay was one hop. An own-cadence loop re-emits the packet every tick from a pass that was itself packet-conditioned, and the program’s own history names the expected failure — drift into a fixed attractor, the way unregularized hypernetworks collapse to input-independence — the creature contemplating nothing, with extra steps. C2 exists to measure exactly that.

2 The ladder and its gates

Three rungs, each registered with its readings before anything ran (`tracks/own-cadence/PLAN.md`), on the creature’s own model — a frozen LFM2.5-230M on Apple-silicon MPS, float32, no GPU borrowed anywhere.

C0 — does the channel exist at 230M? The relay promotion gate transplanted verbatim: nine checks per seed, spanning content (true-packet top-1 ≥ 0.75 restricted to the label bank, shuffled ≤ 0.25 , margin, free generation recovering ≥ 6 of 8 contents, a linear probe on the packet itself), and plumbing (zero-dose bit-exact identity, gradients confined to the bridge, base weights byte-identical, a no-cache assert on the fresh pass-2 context). Eight single-token contents, three seeds, the wrapper discipline holding train/dev/test phrasings disjoint.

C1 — does a goal packet drive action? The armed production stack (the promoted `dream4` self block, treated as a fixed part of the host) plus a goal alphabet 1:1 with the acting menu the creature already parses: `listen_pond`, `propose_experiment`, `diary`, `remember`, `nothing`. Pass 1 is an authored goal-assignment exchange whose text never reaches pass 2; pass 2 is a fresh context with the serve-time action menu, and the score is the parsed tool choice. Gate per seed: goal-consistent choice on held-out phrasings strictly above *three* floors measured in-band — no packet, shuffled (wrong-goal) packet, and `nullin-solo` (is a directed packet better than ignition alone?) — plus the plumbing battery on the armed stack. Chance is 0.20.

C2 — the cadence. Sixteen ticks: each tick, pass 2 acts under the packet; the emitter re-encodes the next packet from that acted pass; nothing else crosses — no text, no cache, plumbing-asserted. Registered curves: retention against a frozen-packet loop (the hop-0 packet replayed every tick, separating carrier decay from re-encoding decay) and a no-packet loop; packet drift (cosine to hop 0 and to the goal-class centroid); and a scripted goal switch at hop 8 with cross-tick contamination counted, not assumed. Registered readings: (i) both arms hold — loop-ready; (ii) re-encoded decays while frozen holds — the emitter is the weak link, and a re-encoding consistency term is the follow-up; (iii) both decay — the carrier leaks; stop and report.

3 Results

C0: the tap was one layer too shallow, and the pair of runs is the finding. At the depth-proportional transplant of the 0.8B relay’s tap (layer 11 of 14, 0.79 depth), the gate fails 6/9 at every seed — and every failure is a content check while every plumbing check stays green (Table 1). The diagnosis is in the probe split: training fits (batch accuracy 0.88–1.00), the probe reads train 0.92–1.00 — and held-out test 0.12–0.19, chance. At layer 11 the packet memorizes phrasing identity, not content. A per-layer linear probe on the same 72 examples (the post-stop diagnostic, quoted from the track record) finds held-out accuracy at or near chance through layer 11 and 1.000 at layers 12 and 13: phrasing-invariant content exists at 230M, and it crystallizes

	tap 11 (registered transplant)			tap 12 (amended)			
seed	top-1	probe tr→te	checks	top-1	probe te	gen	checks
42	0.12	0.94→0.19	6/9	1.00	1.00	1.00	9/9
1042	0.25	0.92→0.19	6/9	0.94	1.00	0.94	9/9
2042	0.19	1.00→0.12	6/9	1.00	1.00	1.00	9/9

Table 1: C0 at 230M: one layer of tap depth is the difference between a channel that memorizes phrasings and one that carries content. Every tap-11 failure is a content check; the six green checks per seed are the plumbing battery.

seed	goal-consistent	no packet	nullin-solo	wrong goal	passed
42	0.90	0.20	0.20	0.40	yes
1042	0.70	0.20	0.20	0.30	yes
2042	0.90	0.20	0.20	0.50	yes

Table 2: C1 at 230M: goal-consistent action choice on held-out phrasings vs the three registered floors, armed production stack, chance 0.20.

exactly at layer 12, the last full-attention layer. The registered amendment moved the tap there and reran everything unchanged: **9 of 9 checks at all three seeds**, 62 seconds wall. At the bottom of the scale ladder the depth-proportional rule breaks — the invariant representation lives only in the last two layers, where the 0.8B had it by 0.79 depth — and the probe scan, not the ratio, is the tap-finder for every future small host.

C1: the goal packet drives the action. All three seeds pass every check (Table 2): goal-consistent choice 0.90/0.70/0.90 against floors that sit where the registration predicted — no-packet at 0.20 (the creature’s habitual pick is chance-level against a rotating goal), **nullin-solo** at 0.20 (undirected waking buys nothing on goal-consistency), shuffled-goal at 0.30–0.50. The shuffled cell is the sharpest secondary reading: a wrong-goal packet still elicits purposeful action-taking — it just aims it wrong. Waking and directing are separable, and only the true packet supplies direction.

C2: reading (ii), unanimous — the carrier holds, the emitter leaks. The frozen-packet arm is goal-consistent at 0.80–1.00 and *flat across all sixteen ticks* (Figure 1, Table 3) — the first measurement of a cycled packet’s lifetime, and the carrier does not decay: written once, the goal directs action indefinitely. The live re-emitted arm shares hop 0 with it (0.87 pooled — the same assignment-written packet) and then loses the goal on the very first re-encoding: 0.20 at hop 1, chance from there on (segment means 0.35–0.40 over hops 0–3, 0.20 over hops 4–7). The failure is located, not mysterious: the packet the emitter writes from an *acted* pass — the action menu plus a JSON tool call, nothing like the assignment exchanges it was trained on — drops to 0.21 cosine against the hop-0 packet at the first re-encoding and stays there (Figure 1, right), so the loop is off-distribution after one hop. The scripted hop-8 switch replays the mechanism in miniature: the fresh assignment-written packet for the new goal is consistent for exactly one tick (0.80 pooled at hop 8), the next re-encoding erases it, and the post-switch segment lands at 0.25–0.30 with cross-tick contamination — the pre-switch goal firing after the switch — at 7–8 of 40 cells, measured, not assumed. 544 seconds wall.

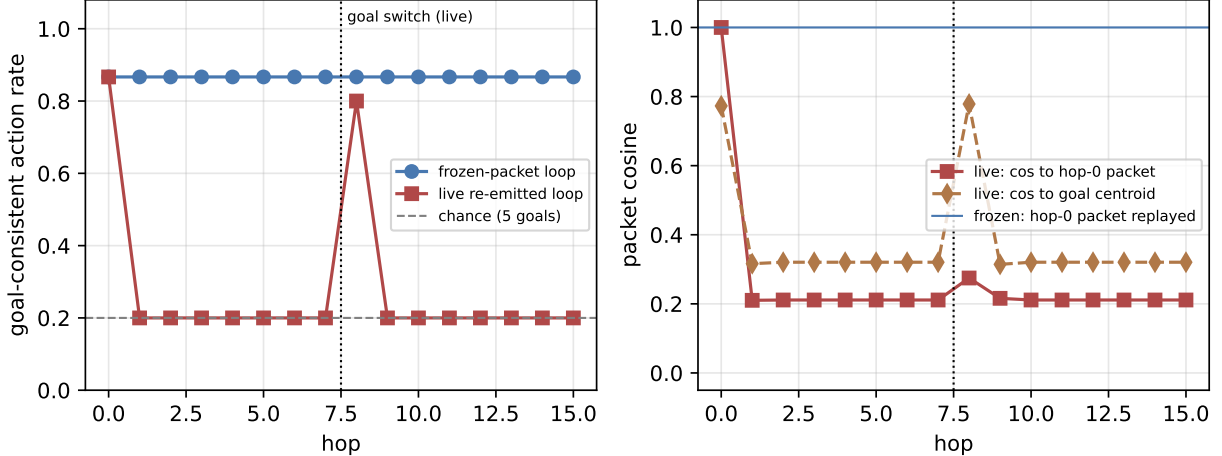


Figure 1: The cadence, pooled over five goals and three seeds. Left: goal-consistent action rate per hop; the frozen-packet loop holds flat for sixteen ticks while the live re-emitted loop collapses to chance on its first re-encoding; the dotted line marks the scripted goal switch in the live arm, whose fresh assignment-written packet buys exactly one consistent tick. Right: the live arm’s packet cosine to the hop-0 packet and to its goal-class centroid — one re-encoding of an acted pass leaves the training distribution, and only the switch’s fresh packet briefly returns.

seed	frozen packet		live re-emitted			
	hops 0–3	hops 12–15	hops 0–3	hops 4–7	post-switch	contam.
42	1.00	1.00	0.40	0.20	0.28	7/40
1042	0.80	0.80	0.35	0.20	0.25	8/40
2042	0.80	0.80	0.35	0.20	0.30	7/40

Table 3: C2 at 230M: goal-consistent rate by segment. The frozen arm does not decay; the live arm’s early segment owes its mean entirely to the shared hop 0 and sits at chance before the switch; contamination counts pre-switch-goal firings in the 40 post-switch cells.

4 What this licenses, and what it names

Three claims stand. The goal-carrier channel exists at creature scale, with a scale lesson about where to tap it. A 256-float packet, armed on the production stack, turns undirected waking into directed action — and the wrong-goal control shows those are different quantities. And a goal written once endures: the frozen arm’s sixteen flat ticks are the carrier’s lifetime measurement, the number the near-term architecture runs on.

That architecture is buildable now, from parts that all passed: a per-goal packet store (write each goal’s packet once, from an assignment exchange — C0/C1’s regime) plus the scripted update rule choosing *which* packet to arm each tick, with interoception driving the rule. Nothing in it asks the emitter to read an acted pass. What does ask for that — the creature *re-writing its own intention* — is now a narrow, named gap rather than an open question: the emitter is trained on assignment surfaces and is off-distribution on acted ones, so the registered follow-up (C2b) adds a re-encoding consistency term — train the emitter so that encoding an acted-under- G pass reproduces the packet for G — and re-runs C2 verbatim. Wiring any of this into the live creature

remains explicitly held for a human decision, per the program’s standing rule; past that, the same presence-mode reasoning that let the goal survive paper 10’s boundary sketches the longer line — goals pursued *through tools*, where the packet carries the want and the context carries the world — each rung to be registered before it runs.

5 Reproducibility

Artifacts `own-cadence-c0-230m{-tap12}` and `own-cadence-c{1,2}-230m` are bundled beside this note. Every number and the figure render from them alone; the one exception is the per-layer tap scan, quoted from the track record (`tracks/own-cadence/PLAN.md`, where every rung’s registration and readings are dated before its run). Harness: `tracks/own-cadence/own_cadence.py`. Host model LiquidAI/LFM2.5-230M at revision 13a53837c490, MPS float32.

References

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